Volatility models for REMX:
GARCH, GARCH-X, GARCHAND, GARCHND and EGARCH-X

Summary of all models estimated in the notebooks **03-GP-models-gaussian**, **03-GP-models-student** and **04-extension-models**. Each section contains the variance expression, a brief description of what the model is meant to demonstrate, and the main estimation results (Gaussian first, Student-t second). The *Sig.* column summarises significance: *** = 1%, ** = 5%, * = 10%.

1. GARCH(1,1) — Baseline

Conditional variance:

$$\sigma^2_t = \omega + \alpha \cdot r^2_{t-1} + \beta \cdot \sigma^2_{t-1}$$

What we want to demonstrate: Canonical Bollerslev (1986) specification. It serves as the benchmark against which every sentiment-augmented model is compared. The goal is to confirm persistence ($\alpha+\beta$ close to 1) and that the basic parameters are significant; all later extensions are evaluated relative to this fit.

Gaussian

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
mu	-0.0062	0.0370	-0.1679	0.8666	
omega	0.0857	0.0301	2.8495	0.0044	***
alpha[1]	0.0892	0.0161	5.5416	0.0000	***
beta[1]	0.8975	0.0183	49.1353	0.0000	***

```
Log-likelihood : -5982.2557
AIC : 11972.5114
BIC : 11996.2106
alpha[1] + beta[1] = 0.9867 (stationary (<1))
```

Student-t

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
mu	-0.0108	0.0361	-0.2983	0.7655	
omega	0.0876	0.0299	2.9323	0.0034	***
alpha[1]	0.0841	0.0150	5.5857	0.0000	***
beta[1]	0.9010	0.0178	50.7265	0.0000	***
nu	11.1692	2.3282	4.7974	0.0000	***

```
Log-likelihood : -5956.1313
AIC : 11922.2626
BIC : 11951.8866
alpha[1] + beta[1] = 0.9851 (stationary (<1))
nu = 11.1692 (smaller -> heavier tails; Gaussian as nu -> infinity)
```

2. GARCH-X (tone) — G&P Eq. (4)

Conditional variance:

$$\sigma^2_t = \omega + \alpha \cdot r^2_{t-1} + \beta \cdot \sigma^2_{t-1} + \gamma \cdot \text{tone}^2_{t-1}$$

What we want to demonstrate: Symmetric extension that injects the daily average news tone as an exogenous regressor in the variance equation. We want to test whether news intensity (regardless of sign) generates additional volatility in REMX, i.e. whether γ is statistically significant.

Gaussian

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0902	0.0482	1.8712	0.0613	*
alpha	0.0902	0.0159	5.6679	0.0000	***
beta	0.8961	0.0181	49.5953	0.0000	***
gamma	-0.0019	0.0227	-0.0842	0.9329	

```
Log-likelihood : -5982.4934
AIC : 11972.9868
BIC : 11996.6859
alpha + beta : 0.9863 (stationary (<1))
```

Student-t

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.1280	0.0479	2.6733	0.0075	***
alpha	0.0845	0.0155	5.4422	0.0000	***
beta	0.8977	0.0194	46.2268	0.0000	***
gamma	-0.0220	0.0145	-1.5235	0.1276	
nu	10.9018	2.2471	4.8515	0.0000	***

```
Log-likelihood : -5954.8048
AIC : 11919.6097
BIC : 11949.2336
alpha + beta : 0.9822 (stationary (<1))
Estimated degrees of freedom: nu_hat = 10.902
Note: t-statistic on nu vs 0 is not economically meaningful (nu > 2 by construction).
```

3. GARCH-X (article growth) — G&P Eq. (4)

Conditional variance:

$$\sigma^2_t = \omega + \alpha \cdot r^2_{t-1} + \beta \cdot \sigma^2_{t-1} + \gamma \cdot \text{art\_growth}^2_{t-1}$$

What we want to demonstrate: Same structure as the previous GARCH-X, but using the daily growth in article volume as a proxy for news intensity. We want to see whether spikes in media coverage (a significant γ) anticipate next-day volatility.

Gaussian

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0919	0.0304	3.0270	0.0025	***
alpha	0.0901	0.0155	5.8108	0.0000	***
beta	0.8955	0.0176	50.7544	0.0000	***
gamma	-0.0007	0.0002	-4.2548	0.0000	***

```
Log-likelihood : -5980.9476
AIC : 11969.8952
BIC : 11993.5944
alpha + beta : 0.9857 (stationary (<1))
```

Student-t

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0935	0.0301	3.1059	0.0019	***
alpha	0.0852	0.0145	5.8845	0.0000	***
beta	0.8989	0.0171	52.4583	0.0000	***
gamma	-0.0006	0.0002	-3.9035	0.0001	***
nu	11.1789	2.3363	4.7848	0.0000	***

```
Log-likelihood : -5954.9782
AIC : 11919.9565
BIC : 11949.5805
alpha + beta : 0.9841 (stationary (<1))
Estimated degrees of freedom: nu_hat = 11.179
Note: t-statistic on nu vs 0 is not economically meaningful (nu > 2 by construction).
```

4. GARCHAND (tone) — G&P Eq. (5)

Conditional variance:

$$\sigma^2_t = \omega + \alpha \cdot r^2_{t-1} + \beta \cdot \sigma^2_{t-1} + \gamma\,(1 + d_1 \cdot \theta) \cdot tone^2_{t-1},\ d_1 = 1\{tone_{t-1} < 0\}$$

What we want to demonstrate: Asymmetric version of GARCH-X: when the previous day's tone is negative, its impact on variance is amplified by the factor (1 + θ). We want to demonstrate that negative news drives a stronger volatility response than positive news (θ > 0 and significant).

Gaussian

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0959	0.0903	1.0617	0.2884	
alpha	0.0900	0.0162	5.5712	0.0000	***
beta	0.8959	0.0191	47.0074	0.0000	***
gamma	-0.0037	0.0185	-0.2017	0.8402	
theta	0.4984	6.6265	0.0752	0.9400	

```
Log-likelihood : -5982.3901
AIC : 11974.7801
BIC : 12004.4041
alpha + beta : 0.9859 (stationary (<1))
Fraction of days with tone_{t-1} < 0 (d1 = 1): 0.481
```

Student-t

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.1309	0.0481	2.7207	0.0065	***
alpha	0.0840	0.0157	5.3364	0.0000	***
beta	0.8969	0.0200	44.7611	0.0000	***
gamma	-0.0110	0.0054	-2.0282	0.0425	**
theta	1.1917	0.8303	1.4354	0.1512	
nu	11.0668	2.3786	4.6527	0.0000	***

```
Log-likelihood : -5954.3046
AIC : 11920.6092
BIC : 11956.1580
alpha + beta : 0.9809 (stationary (<1))
Fraction of days with tone_{t-1} < 0 (d1 = 1): 0.481
Estimated degrees of freedom: nu_hat = 11.067
```

5. GARCHAND (article growth) — G&P Eq. (6)

Conditional variance:

$$\sigma^2_t = \omega + \alpha \cdot r^2_{t-1} + \beta \cdot \sigma^2_{t-1} + \gamma \cdot d_2 \cdot \text{art\_growth}^2_{t-1},\ d_2 = 1\{\text{art\_growth}_{t-1} > 0\}$$

What we want to demonstrate: The article-volume effect is switched on only on days in which the news flow GROWS relative to the previous day. We want to show that novelty (rising coverage) is what moves volatility, rather than low news volumes.

Gaussian

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0872	0.0302	2.8894	0.0039	***
alpha	0.0904	0.0162	5.5778	0.0000	***
beta	0.8961	0.0182	49.2791	0.0000	***
gamma	0.0000	0.0034	0.0000	1.0000	

```
Log-likelihood : -5982.5068
AIC : 11973.0135
BIC : 11996.7127
alpha + beta : 0.9865 (stationary (<1))
Fraction of days with art_growth_{t-1} > 0 (d2 = 1): 0.315
```

Student-t

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0892	0.0306	2.9176	0.0035	***
alpha	0.0852	0.0173	4.9326	0.0000	***
beta	0.8996	0.0190	47.4275	0.0000	***
gamma	0.0000	0.0078	0.0000	1.0000	
nu	11.1153	2.4894	4.4650	0.0000	***

```
Log-likelihood : -5956.2584
AIC : 11922.5168
BIC : 11952.1408
alpha + beta : 0.9848 (stationary (<1))
Fraction of days with art_growth_{t-1} > 0 (d2 = 1): 0.315
Estimated degrees of freedom: nu_hat = 11.115
```

6. GARCHND (tone & article growth) — G&P Eq. (7)

Conditional variance:

$$\sigma^2_t = \omega + \alpha \cdot r^2_{t-1} + \beta \cdot \sigma^2_{t-1} + \gamma \cdot d_3 \cdot x^2_{t-1},\ d_3 = 1\{\sigma^2_{t-1} \geq \kappa\},\ x \in \{\text{tone, art\_growth}\}$$

What we want to demonstrate: The news impact activates only when the previous day's variance exceeds a threshold κ, calibrated from an annualised volatility level (30% and 50%). We want to test whether the sentiment effect is non-linear in the volatility regime: a significant γ in the high-volatility regime would imply that news amplifies volatility only when the market is already stressed.

Gaussian

GARCHND x = Tone kappa = 30% annual

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.1029	0.0308	3.3377	0.0008	***
alpha	0.0904	0.0156	5.8029	0.0000	***
beta	0.8861	0.0154	57.6232	0.0000	***
gamma	0.0441	0.0856	0.5152	0.6064	

```
Log-likelihood : -5979.9708
AIC : 11967.9417
BIC : 11991.6408
alpha + beta : 0.9765 (stationary (<1))
d3 active in-sample (sigma^2_{t-1} >= kappa): 0.622
```

GARCHND x = Tone kappa = 50% annual

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0622	0.0135	4.6004	0.0000	***
alpha	0.0885	0.0079	11.2061	0.0000	***
beta	0.9060	0.0059	153.6288	0.0000	***
gamma	-0.2887	0.0698	-4.1373	0.0000	***

```
Log-likelihood : -5979.9431
AIC : 11967.8861
BIC : 11991.5853
alpha + beta : 0.9945 (stationary (<1))
d3 active in-sample (sigma^2_{t-1} >= kappa): 0.077
```

GARCHND x = Art_growth kappa = 30% annual

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.1010	0.0316	3.1953	0.0014	***
alpha	0.0942	0.0153	6.1488	0.0000	***
beta	0.8862	0.0189	46.8293	0.0000	***
gamma	0.1123	0.0901	1.2456	0.2129	

```
Log-likelihood : -5981.8144
AIC : 11971.6288
```

```
BIC : 11995.3280
alpha + beta : 0.9804 (stationary (<1))
d3 active in-sample (sigma^2_{t-1} >= kappa): 0.620
```

GARCHND x = Art_growth kappa = 50% annual

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0836	0.0241	3.4715	0.0005	***
alpha	0.0904	0.0126	7.1843	0.0000	***
beta	0.8973	0.0135	66.5051	0.0000	***
gamma	-0.0979	0.0932	-1.0498	0.2938	

```
Log-likelihood : -5982.0948
AIC : 11972.1896
BIC : 11995.8888
alpha + beta : 0.9877 (stationary (<1))
d3 active in-sample (sigma^2_{t-1} >= kappa): 0.086
```

Student-t

GARCHND x = Tone kappa = 30% annual

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0896	0.0304	2.9467	0.0032	***
alpha	0.0852	0.0148	5.7543	0.0000	***
beta	0.8994	0.0177	50.6996	0.0000	***
gamma	0.0012	0.0209	0.0571	0.9545	
nu	11.1313	2.2460	4.9559	0.0000	***

```
Log-likelihood : -5956.2566
AIC : 11922.5133
BIC : 11952.1372
alpha + beta : 0.9845 (stationary (<1))
d3 active in-sample (sigma^2_{t-1} >= kappa): 0.611
Estimated degrees of freedom: nu_hat = 11.131
```

GARCHND x = Tone kappa = 50% annual

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0640	0.0149	4.3094	0.0000	***
alpha	0.0842	0.0113	7.4619	0.0000	***
beta	0.9089	0.0088	103.5564	0.0000	***
gamma	-0.2622	0.0953	-2.7499	0.0060	***
nu	11.0950	2.2467	4.9383	0.0000	***

```
Log-likelihood : -5954.5058
AIC : 11919.0116
BIC : 11948.6356
```

```
alpha + beta : 0.9931 (stationary (<1))
d3 active in-sample (sigma^2_{t-1} >= kappa): 0.073
Estimated degrees of freedom: nu_hat = 11.095
```

GARCHND x = Art_growth kappa = 30% annual

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0787	0.0229	3.4356	0.0006	***
alpha	0.0759	0.0066	11.4705	0.0000	***
beta	0.9149	0.0054	169.8811	0.0000	***
gamma	-0.0679	0.0588	-1.1544	0.2483	
nu	10.3328	1.9677	5.2512	0.0000	***

```
Log-likelihood : -5955.9168
AIC : 11921.8335
BIC : 11951.4575
alpha + beta : 0.9907 (stationary (<1))
d3 active in-sample (sigma^2_{t-1} >= kappa): 0.628
Estimated degrees of freedom: nu_hat = 10.333
```

GARCHND x = Art_growth kappa = 50% annual

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
omega	0.0849	0.0252	3.3678	0.0008	***
alpha	0.0851	0.0150	5.6781	0.0000	***
beta	0.9011	0.0167	54.0287	0.0000	***
gamma	-0.0813	0.0929	-0.8756	0.3812	
nu	11.1089	2.2698	4.8943	0.0000	***

```
Log-likelihood : -5956.0217
AIC : 11922.0434
BIC : 11951.6674
alpha + beta : 0.9863 (stationary (<1))
d3 active in-sample (sigma^2_{t-1} >= kappa): 0.081
Estimated degrees of freedom: nu_hat = 11.109
```

7. EGARCH(1,1)-X — Extension model

Conditional variance:

$$\ln(\sigma^2_t) = \omega + \alpha\,[|z_{t-1}| - E|z_{t-1}|] + \xi \cdot z_{t-1} + \beta \cdot \ln(\sigma^2_{t-1}) + \gamma_1 \cdot \text{tone}_{t-1} + \gamma_2 \cdot \text{art\_growth}_{t-1}$$

What we want to demonstrate: Log-variance specification (Nelson, 1991) with two exogenous regressors. It (i) guarantees positivity of σ^2 without parameter restrictions, (ii) captures asymmetry through the leverage term ξ (negative shocks weigh more than positive ones), and (iii) allows tone and art_growth to enter linearly (not as squares). We want to show that EGARCH-X captures asymmetry more cleanly than GARCHAND, and to test whether γ_1 and γ_2 are significant in log-variance rather than in squared variance.

Gaussian

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
mu	-0.0198	0.0369	-0.5379	0.5907	
omega	0.0355	0.0116	3.0514	0.0023	***
alpha	0.1695	0.0283	5.9802	0.0000	***
xi	-0.0230	0.0136	-1.6905	0.0909	*
beta	0.9784	0.0073	134.4085	0.0000	***
gamma_tone	0.0062	0.0062	1.0074	0.3138	
gamma_artgrowth	-0.0159	0.0086	-1.8460	0.0649	*

```
Log-likelihood : -5977.3101
AIC : 11968.6203
BIC : 12010.0939
beta = 0.9784 (stationary (|beta|<1))
Half-life of a log-variance shock: 31.7 trading days
```

Student-t

Parameter	Estimate	Std. Err.	t-ratio	p-value	Sig.
mu	-0.0155	0.0367	-0.4210	0.6737	
omega	0.0316	0.0099	3.2021	0.0014	***
alpha	0.1541	0.0243	6.3453	0.0000	***
xi	-0.0148	0.0108	-1.3785	0.1681	
beta	0.9785	0.0063	154.2750	0.0000	***
gamma_tone	0.0104	0.0049	2.1231	0.0337	**
gamma_artgrowth	-0.0172	0.0098	-1.7493	0.0802	*
nu	10.8176	2.1228	5.0959	0.0000	***

```
Log-likelihood : -5950.0776
AIC : 11916.1552
BIC : 11963.5536
beta = 0.9785 (stationary (|beta|<1))
Half-life of a log-variance shock: 31.9 trading days
```

nu = 10.8176 (Student-t dof; lower -> heavier tails. Note: the t-ratio for nu tests H0: nu = 0, which is not the meaningful null. To test Gaussian vs Student-t use an LR test against the Gaussian EGARCH-X.)